

# Umer Irfan

Lahore · Pakistan

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## EDUCATION

### Lahore University of Management Sciences

BSc Computer Science · Minor in Computer Engineering

### Pak-Turk Maarif International Schools and Colleges

A-Level: 1A\* 2A · O-Level: 10A\*

Sept 2023 to July 2027

CS GPA **3.707/4.00** · CGPA **3.491/4.00**

June 2023

## RESEARCH INTERESTS

Embedded Systems · Hardware Security · Firmware Security · Computer Architecture · Operating Systems

## HONOURS AND AWARDS

- **1st Place.** *Pakistan Semiconductor Summit Hackathon* 2026. RISC-V RTL verification with Sail and Verilator.
- **Dean's Honour List.** LUMS. Three semesters.
- **Best in Category.** Teknofest 2020 in Gaziantep Turkey.
- **Best Project.** Intel ISEF Pakistan 2019.
- **Team Pakistan.** *International Young Physicists Tournament* in Beijing 2018.

## PUBLICATIONS

### Accepted Publications

- *Motion-Coupled Sensing: Self-Powered State-Change Sensing*
- *Kissan-Dost: Conversational IoT for Precision Agriculture*

Co-author · *IEEE MASS 2026*

Co-author · *IEEE DCOSS-IoT 2026*

### Preprints

- *LLM-Enhanced Wearables for Health Guidance in LMICs*

Co-author · *arXiv:2602.08701*

### Submitted Manuscript

- *PARDA: On-Device Audio Privacy for Smart Glasses*

First author · *IEEE PerCom 2027*

### Manuscripts in Preparation

- *Programmer-in-the-Middle: Stealthy Firmware Implantation via Flash Programmers*
- *On-Chip Behavioral Monitoring for Runtime Hardware Trojan Detection*
- *WatchTower: Transformer-Based Open-Set Network Anomaly Detection*

Joint first author

Joint first author

Joint first author

## SELECTED RESEARCH AND TECHNICAL PROJECTS

### Programmer-in-the-Middle (PITM)

2026 to Present

- Exposed an overlooked **firmware supply-chain gap** by treating flash programming equipment as an untrusted stage after compilation and code review.
- Modified USBasp firmware to perform **AVR reset-vector hijacking** and stub injection while returning clean data during host readback verification.
- **Reverse-engineered ST-Link V2** firmware and ran custom programmer firmware across **three target boards**, demonstrating portability beyond one MCU target.

### On-Chip Program Behavioral Adherence Monitoring

2026 to Present

- Designed a **runtime hardware Trojan detector** for rare-trigger attacks that can evade bounded pre-deployment testing.
- Constructed expected and **Trojan-affected execution-trace datasets** across computational and application workloads.
- Developed a 1D CNN detection pipeline and an **FPGA monitor architecture** for evaluation across processor and cache configurations.

### PixelPacker | FPGA-Accelerated Neural Image Compression

2025 to 2026

- Built edge-cloud neural compression for wildlife camera traps. Distilled and INT8-quantized a depthwise-separable encoder to a **36-fold smaller memory footprint**.
- Developed an **Artix-7 FPGA accelerator** with pipelined DSPs, FIFO buffering, and DDR memory access. Implemented **background-aware latent differencing** on ESP32.

### RIZZ-8 | Custom 8-Bit RISC Processor

2025

- Independently built an **8-bit Harvard CPU** from 74-series ICs on **eight custom PCBs** with a custom ISA and 16-bit address space.
- Wrote a **Python assembler**. Loaded machine code into EEPROM and executed custom programs on physical hardware.

### KeySiphon | Power-Rail Side-Channel Analysis

2025

- Instrumented an ATmega328P RSA implementation for **power-trace capture**. Distinguished secret-dependent exponent-bit operations with **93.10% validation accuracy**.

### WatchTower | Transformer-Based Network Anomaly Detection

2025

- Developed a transformer-based detector with **28 packet-level features** and **PROSER open-set recognition** to detect known and unseen network attacks.
- Optimized INT8 inference with TVM on Raspberry Pi 5. Reached **3.82 ms inference latency** and **93.66% classification accuracy**.

### CARDY | ESP8266-Based Interactive Challenge Platform

2023

- Developed *ESP8266 firmware* with an HTTP server and LittleFS and hardware controls. Designed the **custom PCB**.
- Coordinated assembly and testing of **50 physical devices**. Supported **50 participating teams** with real-time telemetry.

RESEARCH EXPERIENCE

Research Assistant | *SYSNET Lab* at LUMS

May 2024 to Sept 2026

*Dr. Naveed Anwar Bhatti*

- Built an arbitrary **voltage-trace replay bench** for CheckMate with laptop-controlled function generators and amplifiers.
- Built a **5 to 70°C** aluminum **Peltier chamber** to characterize temperature-dependent device performance.
- **Designed and hand-assembled a four-layer nRF52 wristband** for maternal-health temperature monitoring in rural Sindh. Targeted four months per charge.
- Co-developed a **lossless 16-bit thermal codec** using temporal and camera-noise predictors. Files averaged **15% smaller than FFV1** at similar speed. No regression across tested sets. Best case was **3-fold smaller**.
- Built a **stepper-driven stress-test rig** for Motion-Coupled Sensing. The harvester transmitted after each of **10,000** accelerated actuations with no visible mechanical wear.
- Built Kissan-Dost's **multi-hop ESP-NOW mesh** to relay soil and climate measurements to a gateway. The system ran at **two field sites** for 45 days each.
- Co-authored **two accepted IEEE papers** and a health-wearables arXiv preprint.

Research Assistant | *CHASE Lab* at LUMS

2025 to 2026

*Dr. Muhammad Ali Siddiqi*

- Developed **FPGA datapaths** for ML inference and characterized **power side-channel leakage** on embedded platforms.
- Assessed a commercial **Pacemaker (IMD)** with **HackRF** and ESP32. Crafted BLE attempts across security modes extended its observed connection window from roughly **2 to 10.81 seconds**. Measured **1.56–1.93 mJ** per connection.
- Advanced **PixelPacker acceleration**, PITM and transaction-trace Trojan monitoring.

TEACHING EXPERIENCE

EE320 Computer Organization & Assembly Language

*Sole TA · 2026.* Six quizzes. Seven homework sets.

AI624 AI on Edge Devices

*TA · 2026.* 82 graduate students.

ENGG100 Engineering Design Fundamentals

*TA · 2024.* 30 engineering freshmen.

CS370 Operating Systems

*TA · 2026.* 295 students. Kernel and shell assignments.

CS210 Data Structures

*TA · 2025.* Programming assignments and automated grading.

Diploma in Digital Health

*Course Designer · 2026.* IoMT classes for 90+ medical professionals.

SELECTED GRADUATE COURSEWORK

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| <ul style="list-style-type: none"><li>• Internet of Things</li><li>• Applied Cryptography</li><li>• Topics in Computer and Network Security</li></ul> | <ul style="list-style-type: none"><li>• AI on Edge Devices</li><li>• Computer Architecture</li><li>• Distributed Systems</li></ul> |
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PROFESSIONAL EXPERIENCE

AI/ML Engineer | *Call Agent Application*

Aug 2024 to March 2025

- Built a **streaming voice agent** using Node.js and WebSockets. Integrated WebRTC VAD with Whisper and GPT-4.
- Cut response latency to **under four seconds** through audio buffering and pipeline optimization.

DevOps Engineer | *Un.I Social App*

May to July 2024

- Built GitHub Actions workflows for containerized deployment and automated application delivery.
- Configured EC2 infrastructure and NGINX reverse proxies for traffic and load management.

React Native Engineer | *Project X*

Nov 2024 to June 2025

- Developed an Android application for a smart security-camera startup.
- Implemented **IoT device onboarding**, Firebase authentication, cloud messaging and backend API integration.

TECHNICAL SKILLS

Programming & HDL

C · C++ · Python · x86, MIPS & RISC-V Assembly · Verilog · SystemVerilog

Microcontrollers

PIC16F876A/877A · ATmega328P · ATmega32U2/8U2 · ATtiny85/16 · STM32 · TM4C123GH6PM · nRF52832 · ESP32/ESP8266

FPGA & Architecture

Basys 3 (Artix-7 XC7A35T-1CPG236C) · Nexys A7 (Artix-7 XC7A100T/XC7A50T) · RISC-V · CPU & ISA design · Sail · Verilator

Embedded Systems

SPI · I<sup>2</sup>C · UART · BLE · ESP-NOW · LoRa · FreeRTOS

Firmware Security

USBasp · ST-Link V2 (SWD) · Ghidra · Binary rewriting · Readback spoofing

Hardware Instrumentation

Oscilloscope · Function generators · HackRF/SDR · Power-trace capture

PCB & Mechanical Design

KiCad · EasyEDA Pro · Eagle · JLCPCB · PCB etching · SolidWorks · Blender

Edge Inference

PyTorch · TensorFlow Lite Micro · TVM · INT8 quantization · Knowledge distillation · FedProx